

1.9.5

Taraka Abhinav - EE25BTECH11016

September 16, 2025

Question

Find the distance between the points $(0, 2\sqrt{5})$ and $(-2\sqrt{5}, 0)$.

Theoretical Solution

Let the points be $\mathbf{A} = (0, 2\sqrt{5})$ and $\mathbf{B} = (-2\sqrt{5}, 0)$.

$$\mathbf{A} = \begin{pmatrix} 0 \\ 2\sqrt{5} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -2\sqrt{5} \\ 0 \end{pmatrix} \quad (1)$$

The distance between two points is given by

$$d = \|\mathbf{A} - \mathbf{B}\| \quad (2)$$

Theoretical Solution

Substituting values,

$$d = \left\| \begin{pmatrix} 0 \\ 2\sqrt{5} \end{pmatrix} - \begin{pmatrix} -2\sqrt{5} \\ 0 \end{pmatrix} \right\| \quad (3)$$

$$= \left\| \begin{pmatrix} 2\sqrt{5} \\ 2\sqrt{5} \end{pmatrix} \right\| \quad (4)$$

$$= \sqrt{(2\sqrt{5})^2 + (2\sqrt{5})^2} \quad (5)$$

Simplifying,

$$d = \sqrt{20 + 20} \quad (6)$$

$$= \sqrt{40} \quad (7)$$

$$= 2\sqrt{10} \quad (8)$$

$$\implies d = \boxed{2\sqrt{10}} \quad (9)$$

C Code - Distance Formula

```
#include <stdio.h>
#include <stdlib.h>
#include <math.h>

// Function to compute distance between (x1,y1) and (x2,y2)
double find_distance(double x1, double y1, double x2, double y2)
{
    double dx = x2 - x1;
    double dy = y2 - y1;
    return sqrt(dx*dx + dy*dy);
}

int main() {
    double x1 = 0, y1 = 2*sqrt(5);
    double x2 = -2*sqrt(5), y2 = 0;
    double d = find_distance(x1, y1, x2, y2);
    printf(Distance = %.4f\n, d);
    return 0;
}
```

Python + C Code

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt
import math

# Load shared library
lib = ctypes.CDLL('./section.so')

# Define function signature
lib.find_distance.argtypes = [ctypes.c_double, ctypes.c_double,
                               ctypes.c_double, ctypes.c_double]
lib.find_distance.restype = ctypes.c_double

# Points
x1, y1 = 0, 2*math.sqrt(5)
x2, y2 = -2*math.sqrt(5), 0

# Call C function
dist = lib.find_distance(x1, y1, x2, y2)
```

```
# Define points
```

```
A = np.array([x1, y1])
```

```
B = np.array([x2, y2])
```

```
# Plot line AB and points
```

```
plt.plot([A[0], B[0]], [A[1], B[1]], 'k-', label='$AB$')
```

```
plt.scatter([A[0], B[0]], [A[1], B[1]], c=['red', 'blue'])
```

```
# Annotate points
```

```
labels = [fA({x1},{y1:.2f}), fB({x2:.2f},{y2})]
```

```
for label, coord in zip(labels, [A, B]):
```

```
    plt.annotate(label, (coord[0], coord[1]),
```

```
                  textcoords=offset points,
```

```
                  xytext=(10, -10), ha='center')
```

```
# Decorations
plt.legend()
plt.grid(True)
plt.axis('equal')
plt.title(fDistance = {dist:.2f})
plt.savefig(2.png, dpi=150)
plt.show()
```

Python Code

```
import numpy as np
import matplotlib.pyplot as plt

# Define the points
A = np.array([0, 2*np.sqrt(5)])
B = np.array([-2*np.sqrt(5), 0])

# Compute distance
dx = B[0] - A[0]
dy = B[1] - A[1]
distance = np.sqrt(dx**2 + dy**2)
print(fDistance = {distance:.2f})

# Plot line AB
plt.plot([A[0], B[0]], [A[1], B[1]], 'k-',
         label=f'Distance = {distance:.2f}')
```

Python Code

```
# Plot points
plt.scatter([A[0], B[0]], [A[1], B[1]], c=['red', 'blue'])
labels = [f'A(0, 25)', f'B(-25, 0)']
coords = [A, B]

# Annotate points
for label, coord in zip(labels, coords):
    plt.annotate(label,
                 (coord[0], coord[1]),
                 textcoords=offset points,
                 xytext=(10, -10),
                 ha='center')

# Decorations
plt.legend()
plt.grid(True)
plt.axis('equal')
plt.title(Distance between Two Points)
plt.savefig(distance_plot.png, dpi=150)
plt.show()
```

Plot

beamer/figs/Figure_1.png